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Equine Gastroenterology

Evaluation of Owner-Reported Clinical Signs and Fecal Occult Blood Tests as Predictors of Equine Gastric Ulcer Syndrome in Competition Dressage Horses

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ABSTRACT

Background: Equine gastric ulcer syndrome (EGUS) adversely affects the health, welfare, and performance of sports horses, requiring gastroscopy for definitive diagnosis. Owners frequently consider girth aversion as highly suggestive of EGUS.

Objectives: To evaluate whether owner-reported clinical signs or fecal occult blood tests (FOBTs) can help identify horses needing gastroscopy for EGUS diagnosis or monitoring treatment, thereby reducing unnecessary procedures in unaffected horses.

Animals: Eighty competition dressage horses referred by veterinarians for EGUS evaluation.

Methods: Case–control study including questionnaire, clinical examination, gastroscopy, and FOBT. Based on gastroscopic findings, horses were classified as cases (presenting equine squamous gastric disease, equine glandular gastric disease, or both) or controls (no lesions). Nineteen horses underwent a follow-up examination after EGUS treatment. Predictive values of clinical signs and fecal hemoglobin were assessed using the Wilson–Brown method, and group differences were analyzed using Mann–Whitney and Fisher's exact tests.

Results: Fifty-five horses (69%) had EGUS. The most commonly reported clinical signs in this cohort included girth aversion (78%), poor performance (33%), and weight loss (30%), with no differences between groups ($p = 0.44–0.99$). Neither clinical signs nor FOBT had sufficient sensitivity or specificity for EGUS screening. Among the 19 horses reevaluated after treatment, resolution of clinical signs did not reliably predict mucosal healing, with poor agreement between owner-reported clinical response and gastroscopic outcome.

Conclusions and Clinical Importance: In competition dressage horses, owner-reported clinical signs and FOBT results are unreliable predictors of EGUS or treatment outcomes. Gastroscopy remains essential for accurately diagnosing and monitoring EGUS in sports horses.

Abbreviations: 95% CI, 95% confidence intervals; EGGD, equine glandular gastric disease; EGUS, equine gastric ulcer syndrome; ESGD, equine squamous gastric disease; FOBT, fecal occult blood test; IQR, interquartile range; ORs, odds ratios.

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1 | Introduction

Equine gastric ulcer syndrome (EGUS) is a prevalent condition in sports horses, comprising equine squamous gastric disease (ESGD) and equine glandular gastric disease (EGGD), which differ anatomically and pathophysiologically [1, 2]. The condition can compromise equine welfare and performance, necessitating accurate diagnosis [1, 3–4]. Sports horses (defined as those engaged in disciplines other than racing, such as dressage, show jumping, eventing, and endurance) [5] have varying prevalences of EGUS based on discipline, training intensity, and lesion type. Rates of ESGD range from 25% to 65% of show jumpers [6–8], and up to 93% of high-level endurance horses [9], particularly when exposed to intermittent feeding or transport [10–12]. Although EGGD is generally less prevalent than ESGD, with prevalence rates of 51%–57% in sports horses [1, 8, 13], it can occur in the absence of concurrent ESGD, and it is considered to have a distinct etiopathogenesis [7, 14]. The prevalence of EGUS in competition dressage horses is poorly documented.

Various clinical signs have been associated with EGUS in adult horses, including behavioral changes (e.g., aggression, biting, cribbing), reduced appetite, bruxism, weight loss, poor coat condition, poor performance, chronic diarrhea, and recurrent mild colic [15–19], although causality remains unconfirmed [2, 20]. Horses with EGGD often show more severe signs of gastric discomfort than those with ESGD [1]. Some affected Thoroughbred racehorses show no overt signs, particularly in the early stages [14]. Poor performance can be associated with both ESGD and EGGD in racehorses [3, 21], although this relationship lacks specificity and remains multifactorial [22, 23]. Notably, behavioral complaints such as girth aversion (commonly interpreted by owners as indicative of gastric discomfort in sports horses) are not consistently correlated with either ESGD or EGGD in controlled studies [8, 24–25].

Gastroscopy is currently the reference standard for diagnosis [26]. Despite being minimally invasive, many owners remain reluctant to undergo gastroscopy due to the need to withhold food and administer sedation, potential complications, and associated costs [27, 28]. As empiric treatment without gastroscopic confirmation is discouraged [2], alternative screening tools might help identify horses that require gastroscopy. Methods such as sucrose permeability testing, owner questionnaires, and salivary or serum biomarkers have yielded inconclusive diagnostic value [29–32]. The Horse Grimace Scale lacks discriminatory power for EGUS [33], although an association with the Ridden Horse Pain Ethogram [34] has been observed in a small cohort [35]. The fecal occult blood test (FOBT), an immunoassay detecting albumin and hemoglobin in feces, has demonstrated in vitro analytical validity [36]. However, its reliability for detecting upper gastrointestinal lesions has been inconsistent in humans [37, 38], cattle [39], and alpacas [40]. In horses, the results across studies to date, including the manufacturer's, are variable, and none demonstrate sufficient sensitivity or specificity to support clinical use [41–43]. Potential false positives might result from rectal interventions, trauma, or concurrent gastrointestinal disorders such as protein-losing enteropathy [44].

The current study aimed to (1) determine the prevalence of EGUS and the response to treatment in a referral population of Dutch competition dressage horses and (2) assess the diagnostic value of owner-reported clinical signs and FOBT in EGUS screening and monitoring of treatment outcomes in sports horses.

2 | Material and Methods

2.1 | Animals and Study Design

This study was approved by the Animal Ethics Committee of the University of Liège, Belgium (Approval No. 24-2765). Data collection was performed between July 2021 and December 2024 at a veterinary referral center for equine sports medicine (Sportpaardenarts—Equine Sports Medicine, Zeewolde, the Netherlands). Part of the data was collected retrospectively. In routine clinical practice at the center, standardized data (including detailed anamnesis and owner-reported clinical signs) are systematically recorded using a structured questionnaire completed at the time of admission. Owners provide general consent for the anonymized use of clinical data for scientific purposes upon admission. For inclusion in this study, additional informed consent from owners was obtained retrospectively and prospectively, after formal ethics approval for the study.

Horses referred for gastroscopy by veterinary practitioners at the owners' request and actively competing in national or international dressage competitions within the previous 12 months were included. Both horses with and without clinical signs were eligible for inclusion. Horses without clinical signs were included based on owner-reported concerns, often related to established risk factors such as frequent transport or competition [11, 45]. Horses previously treated with antiulcer medication (e.g., omeprazole, misoprostol, and sucralfate) or with a history of bleeding or observable blood around the face or mouth were excluded from the study.

2.2 | Study Design

A case–control study was used. Cases were horses diagnosed with EGUS by gastroscopy. Controls were horses initially suspected of having EGUS, however without gastroscopic evidence of EGUS.

2.3 | Data Collection

Upon arrival, a detailed anamnesis was recorded, which included the horse's physical activity, medical history, medication use, daily exercise, management, and dietary practices. Owners were asked to report any clinical signs observed in the past 6 months, including weight loss, poor performance, behavioral changes, girth aversion, recurrent colic, chronic diarrhea, poor coat condition, and bruxism. Weight loss included the failure to gain weight despite appropriate feeding and training programs. Poor performance was defined as performing below expectations, reluctance to engage in routine activities, or exercise intolerance. Behavioral changes encompassed stereotypies, irritability in various settings (e.g., barn, paddock, pasture, transport), and abnormal social

interactions with other horses. Girth aversion referred to signs of abnormal behavior when saddled or during girth tightening (e.g., ear pinning, tail swishing, biting, pawing, moving away). Recurrent colic was characterized by more than one episode of colic within two consecutive weeks, resolving without veterinary medical or surgical intervention. Chronic diarrhea was identified by softer-than-expected feces, including the regular observation of feces on the hind limbs or tail. Poor coat condition was defined as a dull or coarse coat in the absence of recent dietary changes. Bruxism referred to repetitive clenching or grinding of teeth. Owners' responses and clinical variables, including body condition scores on a 9-point scale [46], were collected by a European Specialist in Veterinary Sports Medicine and Rehabilitation (T.F.) using a standardized form (File S1).

2.4 | Gastroscopy

Feed was withheld for 10–16 h, and water was withdrawn for 4–10 h before gastroscopy. Sedation was achieved with detomidine hydrochloride (Detogesic, Zoetis, Capelle aan den Ijssel, the Netherlands) at 0.06 mg/kg IV 5 min before the procedure. A flexible 3.2 m video-endoscope (VO-320, Optomed, Les Ulis, France) was introduced via the ventral nasal meatus, advancing through the nasal passage, nasopharynx, esophagus, and stomach. Air was insufflated into the stomach through the gastroscope channel to facilitate observation of the gastric mucosa. Lesions of the squamous and glandular gastric mucosa were assessed in real time during gastroscopy by a European Specialist in Veterinary Sports Medicine and Rehabilitation (T.F.), using the 4-point grading systems established by the Equine Gastric Ulcer Council (Table 1) [47, 48], and horses were categorized into four groups:

- ESGD: horses with an ESGD grade ≥ 2/4 and no EGGD lesions;
- EGGD: horses with an EGGD grade ≥ 2/4 and no ESGD lesions;
- ESGD/EGGD: horses with an ESGD grade ≥ 2/4, and an EGGD grade ≥ 2/4;
- Control: horses without evidence of ESGD or EGGD lesions (grade ≤ 1/4).

2.5 | FOBT

Rectal fecal samples were collected from the first 49 horses directly after gastroscopy while the horses were still sedated. A stall-side FOBT (SUCCEED Equine Fecal Blood Test, Freedom Health LLC, Aurora, OH, USA), a qualitative lateral flow immunoassay, was performed according to the manufacturer's instructions (File S2) to detect albumin and hemoglobin in the fecal samples. Briefly, feces were diluted with clean tap water and thoroughly shaken, and two drops were applied to the designated wells for the detection of albumin and hemoglobin. Results were read after 15 min, and the appearance of the Control Line indicated the validity of each test (Figure 1). All tests were performed using the same batch of FOBT kits.

2.6 | Follow-Up

Horses diagnosed with EGUS received pharmaceutical treatment in accordance with the European College of Equine Internal Medicine consensus statement [2], along with appropriate management changes [49]. ESGD cases received omeprazole (GastroGard, Boehringer Ingelheim Animal Health, Amsterdam, The Netherlands) at 4 mg/kg PO once daily for 3 weeks. EGGD and ESGD/EGGD cases received omeprazole at 4 mg/kg PO once daily, combined with sucralfate (magistral preparation, Grovet, Soesterberg, The Netherlands) at 12 mg/kg PO twice daily for 4 weeks. Exercise regimens were maintained. Follow-up gastroscopy findings were included only when follow-up gastroscopy was performed 24 h after the last treatment administration. The owners' perceptions of clinical signs were re-evaluated. The FOBT was not repeated.

2.7 | Statistical Analysis

Data were collected in a Microsoft Excel worksheet (Microsoft Corporation, Redmond, WA, USA). Normality was assessed using the Shapiro–Wilk test. Nonnormally distributed data were described using the median with the corresponding interquartile range (IQR) and range. Comparisons between groups were performed using the Mann–Whitney test for quantitative variables and Fisher's exact test for qualitative variables, with analyses conducted with Prism 10.5.0 for macOS (GraphPad Software, Boston, MA, USA). Statistical significance was set at $p < 0.05$.

TABLE 1 | The 5-point ordinal grading system for scoring squamous and glandular gastric disease according to the Equine Gastric Ulcer Council [47, 48].

Grade	Squamous mucosa	Glandular mucosa
0	The epithelium is intact and there is no appearance of hyperkeratosis	The epithelium is intact and there is no appearance of hyperemia
1	The mucosa is intact, but there are areas of hyperkeratosis	The mucosa is intact, but there are areas of hyperemia
2	Small, single or multifocal lesions	
3	Large single or extensive superficial lesions	
4	Extensive lesions with areas of apparent deep lesions	

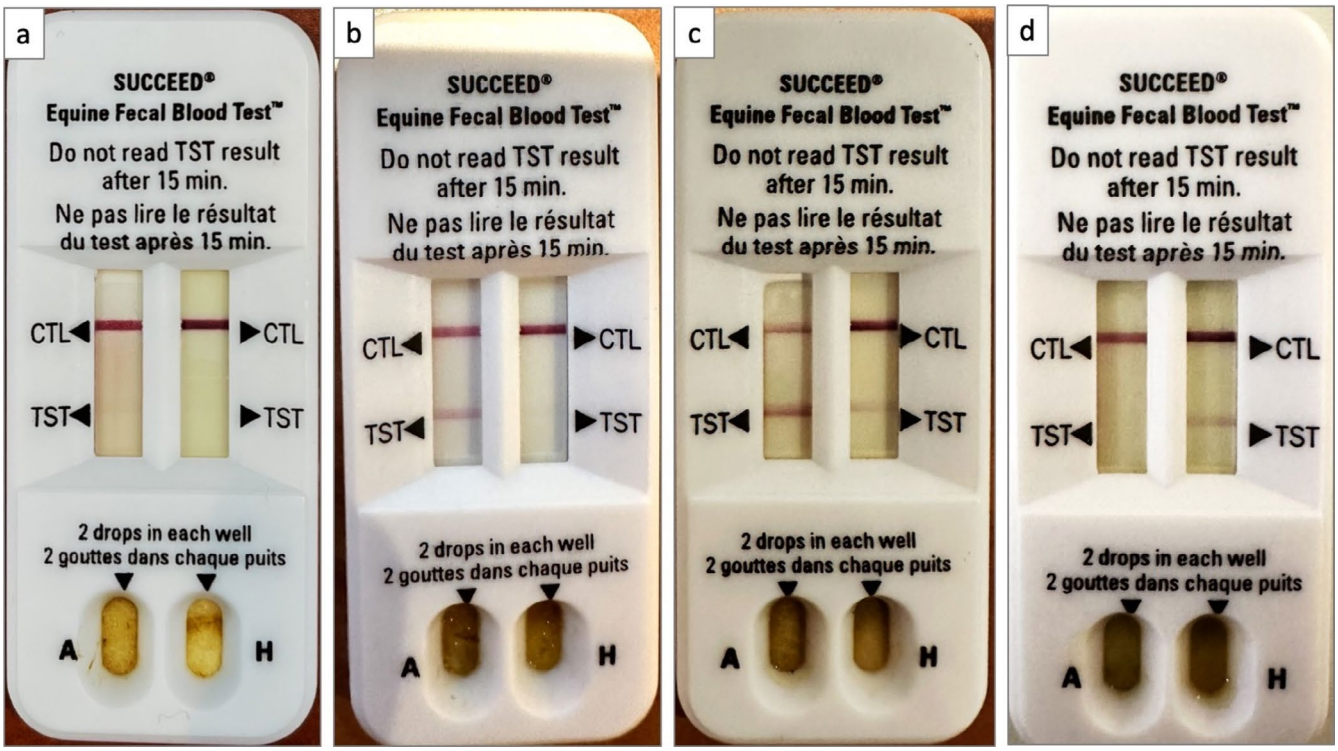

FIGURE 1 | Examples of fecal occult blood test results from four horses. According to the manufacturer, a solid line of any shade represents a positive result. Each test detects albumin (A) and hemoglobin (H), with a control (CTL) line indicating the validity of the protein assessment (TST). (a) Neither albumin nor hemoglobin was detected. (b) Albumin was present, but hemoglobin was not. (c) Both albumin and hemoglobin were detected. (d) Hemoglobin was present, but albumin was not.

Sensitivity and specificity of clinical signs and FOBT results were calculated using the hybrid Wilson–Brown method, while 95% confidence intervals (95% CIs) for the prevalence of each clinical sign or FOBT result were estimated using the Wilson method [50, 51]. The sensitivity (or true positive rate) refers to the proportion of actual positives (horses with EGUS) correctly identified by a clinical sign or a positive FOBT. The specificity (or true negative rate) refers to the proportion of actual negatives (controls) correctly identified by the absence of a clinical sign or a negative FOBT. The formulae for these variables are

$$\text{Sensitivity} = \frac{\text{True positives (TP)}}{\text{TP} + \text{False negatives (FN)}} \quad (1)$$

$$\text{Specificity} = \frac{\text{True negatives (TN)}}{\text{TN} + \text{False positives (FP)}} \quad (2)$$

For the predictive variables, an interpretive range of 0.90–1.00 was considered good to excellent, 0.80–0.89 was acceptable, 0.70–0.79 was moderate, and values below 0.70 were considered poor. Odds ratios (ORs) and 95% CI were computed using the Baptista–Pike method [52].

For horses undergoing follow-up gastroscopy, EGUS status was compared to the initial status using a Bowker's test of symmetry. The distribution of clinical signs and lesion grades was presented using frequency tables and compared to the EGUS status established at follow-up gastroscopy using a Fisher's exact test.

3 | Results

3.1 | Demography and Anamnesis

Eighty horses met the inclusion criteria; none were excluded. The demographic details of the horses in each group are presented in Table 2. Anamnesis data, including management and dietary variables, are stratified by lesion type (ESGD, EGGD, ESGD/EGGD, and total cases) in Table 2. No significant differences were observed between groups.

The cohort comprised 36 geldings (45%), 29 mares (36%), and 15 stallions (19%). Of these, 66 (83%) were Warmblood horses, and 14 (18%) were mixed breeds. The median age of the entire group was 9 years (IQR: 5–12; range: 4–18), with no difference between cases and controls ($p = 0.06$). The median weight of the whole group was 613 kg/1351 lb. (IQR: 577–637 kg/1272–1404 lb.; range: 375–712 kg/827–1570 lb), with no difference between cases and controls ($p = 0.79$). The median body condition score of the entire group was 4.5 (IQR: 4.0–5.0; range: 2.0–7.0), with no difference between cases and controls ($p = 0.49$). Among the 80 horses, 61 (76%) were currently in training (40/55 cases [73%] and 21/25 controls [84%]), undergoing a median of five training sessions per week (IQR: 4–5;

TABLE 2 | Demographics and anamnesis of the 80 competition dressage horses referred for investigation of suspected equine gastric ulcer syndrome, including horses with (cases) or without (controls) equine squamous gastric disease (ESGD) or equine glandular gastric disease (EGGD).

	Cases					
	ESGD ($n=17$)	EGGD ($n=17$)	ESGD/EGGD ($n=21$)	Total cases ($n=55$)	Controls ($n=25$)	p
Type						0.86[b]
Horse	15 (88.2%)	16 (94.1%)	20 (95.2%)	51 (92.7%)	24 (96.0%)	
Pony	2 (11.8%)	1 (5.9%)	1 (4.8%)	4 (7.3%)	1 (4.0%)	
Age						0.13[a]
≤4 years	0 (0.0%)	2 (11.8%)	4 (19.0%)	6 (10.9%)	5 (20.0%)	
5–8 years	6 (35.3%)	3 (17.6%)	8 (38.1%)	17 (30.9%)	10 (40.0%)	
9–12 years	8 (47.1%)	8 (47.1%)	5 (23.8%)	21 (38.2%)	7 (28.0%)	
≥13 years	3 (17.6%)	4 (23.5%)	4 (19.0%)	11 (20.0%)	3 (12.0%)	
Sex						0.72[b]
Gelding	9 (52.9%)	8 (47.1%)	6 (28.6%)	23 (41.8%)	13 (52.0%)	
Mare	6 (35.3%)	6 (35.3%)	9 (42.9%)	21 (38.2%)	8 (32.0%)	
Stallion	2 (11.8%)	3 (17.6%)	6 (28.6%)	11 (20.0%)	4 (16.0%)	
Body condition score						0.87[a]
<4/9	3 (17.6%)	3 (17.6%)	2 (9.5%)	8 (14.5%)	3 (12.0%)	
4–5/9	11 (64.7%)	11 (64.7%)	17 (81.0%)	39 (70.9%)	19 (76.0%)	
>5/9	3 (17.6%)	3 (17.6%)	2 (9.5%)	8 (14.5%)	3 (12.0%)	
Physical activity						
Currently in training						0.24[b]
Yes	15 (88.2%)	11 (64.7%)	14 (66.7%)	40 (72.7%)	21 (84.0%)	
No	2 (11.8%)	6 (35.3%)	7 (33.3%)	15 (27.3%)	4 (16.0%)	
Frequency of training sessions (per week)						0.98[a]
3–4	4 (23.5%)	2 (11.8%)	2 (9.5%)	8 (14.5%)	10 (40.0%)	
5	10 (58.8%)	6 (35.3%)	7 (33.3%)	23 (41.8%)	9 (36.0%)	
6–7	1 (5.9%)	3 (17.6%)	5 (23.8%)	9 (16.4%)	2 (8.0%)	

(Continues)

TABLE 2 | (Continued)

	Cases					
	ESGD ($n=17$)	**EGGD ($n=17$)**	**ESGD/EGGD ($n=21$)**	**Total cases ($n=55$)**	**Controls ($n=25$)**	***p***
Duration of training (per day)						0.07[a]
30 min	6 (35.3%)	5 (29.4%)	6 (28.6%)	17 (30.9%)	9 (36.0%)	
45 min	8 (47.1%)	6 (35.3%)	7 (33.3%)	21 (38.2%)	11 (44.0%)	
1 h	1 (5.9%)	0 (0.0%)	1 (4.8%)	2 (3.6%)	1 (4.0%)	
Currently in competition						0.80[b]
Yes	3 (17.6%)	1 (5.9%)	3 (14.3%)	7 (12.7%)	4 (16.0%)	
No	14 (82.4%)	16 (94.1%)	18 (85.7%)	48 (87.3%)	21 (84.0%)	
Frequency of shows (per month)						0.64[a]
1	2 (11.8%)	0 (0.0%)	1 (4.8%)	3 (5.5%)	3 (12.0%)	
2	0 (0.0%)	1 (5.9%)	2 (9.5%)	3 (5.5%)	1 (4.0%)	
≥ 3	1 (5.9%)	0 (0.0%)	0 (0.0%)	1 (1.8%)	0 (0.0%)	
Environment						
Bedding type						0.82[b]
Straw	16 (94.1%)	15 (88.2%)	17 (81.0%)	48 (87.3%)	19 (76.0%)	
Woodshavings	0 (0.0%)	0 (0.0%)	1 (4.8%)	1 (1.8%)	1 (4.0%)	
Others	1 (5.9%)	2 (11.8%)	3 (14.3%)	6 (10.9%)	5 (20.0%)	
Pasture access						0.45[b]
Yes	16 (94.1%)	14 (82.4%)	20 (95.2%)	50 (90.9%)	24 (96.0%)	
No	1 (5.9%)	3 (17.6%)	1 (4.8%)	5 (9.1%)	1 (4.0%)	
Duration of pasture access (per day)						0.63[b]
< 6 h	11 (64.7%)	9 (52.9%)	14 (66.7%)	34 (61.8%)	13 (52.0%)	
≥ 6 h	5 (29.4%)	5 (29.4%)	6 (28.6%)	16 (29.1%)	8 (32.0%)	
24 h	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (12.0%)	
Feeding strategy (concentrates)						
Feeding concentrates						0.34[a]

(Continues)

TABLE 2 | (Continued)

	Cases					
	ESGD ($n=17$)	**EGGD ($n=17$)**	**ESGD/EGGD ($n=21$)**	**Total cases ($n=55$)**	**Controls ($n=25$)**	***p***
Yes	17 (100.0%)	17 (100.0%)	20 (95.2%)	54 (98.2%)	23 (92.0%)	
No	0 (0.0%)	0 (0.0%)	1 (4.8%)	1 (1.8%)	2 (8.0%)	
Frequency of feeding (per day)						0.34[a]
1	1 (5.9%)	2 (11.8%)	5 (23.8%)	8 (14.5%)	1 (4.0%)	
2	3 (17.6%)	6 (35.3%)	1 (4.8%)	10 (18.2%)	8 (32.0%)	
3	11 (64.7%)	9 (52.9%)	11 (52.4%)	31 (56.4%)	14 (56.0%)	
4–5	2 (11.8%)	0 (0.0%)	3 (14.3%)	5 (9.1%)	0 (0.0%)	
Quantity (per feeding session)						0.64[a]
≤ 250 g	3 (17.6%)	3 (17.6%)	7 (33.3%)	13 (23.6%)	6 (24.0%)	
251–500 g	10 (58.8%)	6 (35.3%)	7 (33.3%)	23 (41.8%)	7 (28.0%)	
501–1000 g	4 (23.5%)	7 (41.2%)	4 (19.0%)	15 (27.3%)	10 (40.0%)	
> 1000 g	0 (0.0%)	1 (5.9%)	2 (9.5%)	3 (5.5%)	0 (0.0%)	
Feeding strategy (roughage)						
Type of roughage						0.64[b]
Hay	10 (58.8%)	12 (70.6%)	11 (52.4%)	33 (60.0%)	15 (60.0%)	
Silage	6 (35.3%)	2 (11.8%)	8 (38.1%)	16 (29.1%)	7 (28.0%)	
Hay/silage combination	1 (5.9%)	3 (17.6%)	2 (9.5%)	6 (10.9%)	3 (12.0%)	
Frequency of feeding (per day)						0.21[a]
2–3	12 (70.6%)	11 (64.7%)	14 (66.7%)	37 (67.3%)	13 (52.0%)	
4–5	1 (5.9%)	2 (11.8%)	1 (4.8%)	4 (7.3%)	5 (20.0%)	
ad libitum	4 (23.5%)	4 (23.5%)	6 (28.6%)	14 (25.5%)	7 (28.0%)	
Quantity (per feeding session)						0.68[a]
< 3 kg	2 (11.8%)	1 (5.9%)	1 (4.8%)	4 (7.3%)	3 (12.0%)	
3–3.9 kg	4 (23.5%)	6 (35.3%)	6 (28.6%)	16 (29.1%)	9 (36.0%)	

(Continues)

TABLE 2 | (Continued)

	Cases					
	ESGD ($n=17$)	**EGGD ($n=17$)**	**ESGD/EGGD ($n=21$)**	**Total cases ($n=55$)**	**Controls ($n=25$)**	***p***
4–8 kg	7 (41.2%)	6 (35.3%)	8 (38.1%)	21 (38.2%)	6 (24.0%)	
ad libitum	4 (23.5%)	4 (23.5%)	6 (28.6%)	14 (25.5%)	7 (28.0%)	
Feeding strategy (others)						
Dietary supplement						0.16[b]
Yes	8 (47.1%)	3 (17.6%)	9 (42.9%)	20 (36.4%)	6 (24.0%)	
No	9 (52.9%)	14 (82.4%)	12 (57.1%)	35 (63.6%)	19 (76.0%)	
Salt block						0.97[b]
Yes	8 (47.1%)	7 (41.2%)	9 (42.9%)	24 (43.6%)	12 (48.0%)	
No	9 (52.9%)	10 (58.8%)	12 (57.1%)	31 (56.4%)	13 (52.0%)	

[a]Significance of differences between cases and controls was assessed using Mann–Whitney test, with $p<0.05$.
[b]Significance of differences between cases and controls was assessed using Fisher's exact test, with $p<0.05$.

range: 3–7), each for a median duration of 45 min (IQR: 30–45; range: 30–60), with no difference between cases and controls ($p>0.99$). The remaining 19 horses (24%) were not in training because their owners deemed them unrideable for various reasons, including orthopedic problems and behavioral changes during riding. Eleven horses (14%) had competed a median of once in the past month (IQR: 1–2; range: 1–4). The median duration of daily confinement was 18 h (IQR: 16–20; range: 0–24), with cases spending a median of 18 h (IQR: 16–20; range: 0–24) confined in their stable, significantly more than 17 h observed in controls (IQR: 8–18; range: 0–21; $p=0.01$). Five cases and one control had no daily access to pasture due to orthopedic reasons.

Sixty-seven horses (84%) were housed on straw bedding, while two (3%) were on wood shavings and 11 (14%) on other types of bedding, with no difference between groups ($p=0.30$). Forty-eight horses (60%) were fed hay, while 23 (29%) received silage and nine (11%) received a combination of both, with no difference between groups ($p>0.99$). Twenty-one horses (26%) received roughage ad libitum, including 14 cases (25%) and 7 controls (28%). The median feeding frequency for roughage to the remaining 59 horses (74%) was three times (IQR: 3–3; range: 2–5) per day, with no difference between groups ($p=0.06$). Three horses (3.8%) did not receive concentrates. For the remaining 77 horses (96%), the median feeding frequency for concentrates was 3 times (IQR: 2–3; range: 1–5) per day, with no difference between groups ($p=0.58$). Most horses (55%) had no access to a salt block, with no difference between groups ($p=0.81$). Most horses (69%) did not receive a nutritional supplement, with no difference between groups ($p=0.82$). One case (2%) and one control (4%) had received nonsteroidal anti-inflammatory drugs. Three EGGD cases (18%) had received corticosteroid treatment within the past 2 weeks, whereas none of the other cases or controls had been treated ($p=0.02$).

3.2 | Gastroscopic Findings

3.2.1 | At Presentation

Gastroscopy was successfully performed in all horses without complications. Despite owners reporting clinical signs suggestive of EGUS, 25 horses (31%) had no evidence of ulcers, while 55 horses (69%) were cases. Among the 71 horses with owner-reported clinical signs suggestive of EGUS, 24 (34%) had no gastroscopic evidence of ulcers, while 47 (66%) were diagnosed with EGUS. Of the nine horses in which the owners had not observed clinical signs but perceived them to be at risk, eight (89%) had EGUS.

From the entire studied population ($n=80$), 38 (48%) had squamous lesions with a median ESGD lesion grade of 3 (IQR: 2–3; range: 2–4), with no difference in ESGD grade between the 17 ESGD cases and the 21 ESGD/EGGD cases ($p=0.28$; Table 3). The ESGD lesions were primarily located adjacent to the *margo plicatus*, with 36 horses (95%) having lesions along the greater curvature of the stomach, and 12 horses (32%) along the lesser curvature. Thirty-eight of 80 horses (48%) had glandular lesions, with a median EGGD lesion grade of 3 (IQR: 2–3; range: 2–4). The 21 horses with combined ESGD/EGGD had a higher EGGD lesion grade, with a median score of 3 (IQR: 3–3; range 2–4)

TABLE 3 | Findings of the gastroscopic examinations at presentation (before) in 55 competition dressage horses with equine gastric ulcer syndrome, including equine squamous gastric disease (ESGD) and equine glandular gastric disease (EGGD), and at follow-up (after) in 19 cases.

	ESGD		EGGD		ESGD/EGGD		Total cases	
	Before ($n=17$)	**After ($n=3$)**	**Before ($n=17$)**	**After ($n=6$)**	**Before ($n=21$)**	**After ($n=10$)**	**Before ($n=55$)**	**After ($n=19$)**
ESGD grade (/4)								
≤1	—	2 (66.7%)	17 (100.0%)	6 (100.0%)	—	7 (70.0%)	17 (30.9%)	15 (78.9%)
2	9 (52.9%)	0 (0.0%)	—	—	8 (38.1%)	3 (30.0%)	17 (30.9%)	3 (15.8%)
3	6 (35.3%)	0 (0.0%)	—	—	7 (33.3%)	0 (0.0%)	13 (23.6%)	0 (0.0%)
4	2 (11.8%)	1 (33.3%)	—	—	6 (28.6%)	0 (0.0%)	8 (14.5%)	1 (5.3%)
EGGD grade (/4)								
≤1	17 (100.0%)	4 (100.0%)	—	2 (33.3%)	—	3 (30.0%)	17 (30.9%)	9 (47.4%)
2	—	—	10 (58.8%)	0 (0.0%)	4 (19.0%)	2 (20.0%)	14 (25.5%)	2 (10.5%)
3	—	—	5 (29.4%)	4 (66.7%)	15 (71.4%)	5 (50.0%)	20 (36.4%)	9 (47.4%)
4	—	—	2 (11.8%)	0 (0.0%)	2 (9.5%)	0 (0.0%)	4 (7.3%)	0 (0.0%)
Gastric parasites								
Absent	16 (94.1%)	3 (100.0%)	15 (88.2%)	6 (100.0%)	15 (71.4%)	10 (100.0%)	44 (80.0%)	19 (100.0%)
Present	1 (5.9%)	0 (0.0%)	2 (11.8%)	0 (0.0%)	6 (28.6%)	0 (0.0%)	11 (20.0%)	0 (0.0%)

compared to a median score of 2 (IQR: 2–3; range 2–4) in the 17 horses with EGGD alone ($p = 0.04$). All EGGD lesions were found adjacent to the pylorus. No abnormalities were detected in the duodenum or esophagus. Eleven horses (14%) had larvae of *Gasterophilus* spp., with no difference between cases and controls ($p = 0.21$).

3.2.2 | After Treatment of EGUS

Only 19 of the 55 cases (35%) were examined at follow-up, including 3 ESGD, 6 EGGD, and 10 ESGD/EGGD cases (Table 3), with no difference in the distribution of gastroscopic findings compared to the initial presentation ($p = 0.06$). The remaining 36 horses (66%) were lost to follow-up for various reasons: owners did not request follow-up because the horses showed clinical improvement ($n = 19$), follow-up examinations were not conducted directly after treatment ($n = 11$), financial constraints ($n = 4$), or no reason was provided ($n = 2$).

Seven of the 19 cases with follow-up examination (37%) fully recovered from EGUS. Two of the three ESGD cases (67%) and seven of the 10 ESGD/EGGD cases (70%) had complete healing of the squamous lesions (ESGD grade ≤ 1). Two of the follow-up horses (11%) had an improvement in their ESGD grade, one (5%) had no change, and one (5%) had a worsening of the ESGD grade. Two of the six EGGD cases (33%) and three of the 10 ESGD/EGGD cases (30%) had complete healing of the glandular lesions (EGGD grade ≤ 1). Five of the follow-up horses (31%) had an improvement in their EGGD grade, five (31%) had no change, and one (6%) had a worsening of the EGGD grade.

Neither of the two horses that had larvae of *Gasterophilus* spp. on the gastric mucosa before treatment had larvae detected after treatment.

3.3 | Value of Owner-Reported Clinical Signs

3.3.1 | For Screening of EGUS

The frequency of clinical signs reported by the owners is presented in Figure 2. These include girth aversion (78%; 95% CI: 67%–85%), poor performance (33%; 95% CI: 23%–43%), weight loss (30%; 95% CI: 21%–41%), behavioral changes (28%; 95% CI: 19%–38%), poor coat condition (23%; 95% CI: 15%–33%), chronic diarrhea (21%; 95% CI: 14%–31%), bruxism (19%; 95% CI: 12%–29%), and recurrent colic (13%; 95% CI: 7%–22%). The owners reported no other clinical signs. Girth aversion was the most frequently reported clinical sign across all groups: 71% (95% CI: 47%–87%) of ESGD, 82% (95% CI: 59%–94%) of EGGD, 76% (95% CI: 55%–89%) of ESGD/EGGD, and 80% (95% CI: 64%–86%) of controls (Table 4). According to their owners, nine horses showed no suggestive clinical signs, including eight of the 55 cases (15%) and one of the 25 controls (4%; $p = 0.26$).

The predictive values are presented in Table 4. Girth aversion had moderate sensitivity for ESGD and ESGD/EGGD, and acceptable sensitivity for EGGD, but poor specificity for all conditions. Poor coat condition and recurrent colic had poor sensitivity for ESGD, EGGD, ESGD/EGGD, and overall (all cases), but acceptable specificity. Weight loss, chronic diarrhea, and bruxism had poor sensitivity across all disease classifications, but had moderate specificity. Poor performance and behavioral

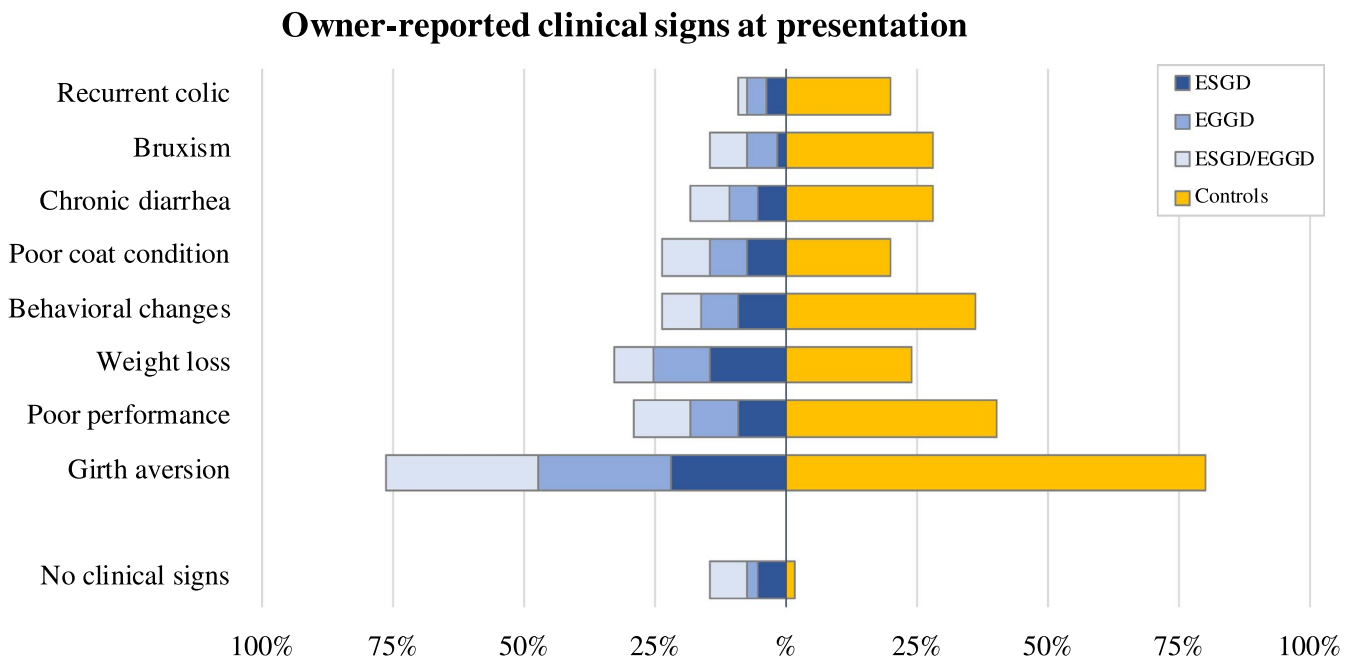

FIGURE 2 | Frequency of owner-reported clinical signs at presentation in 80 competition dressage horses with suspected equine gastric ulcer syndrome (EGUS). Among these, 55 horses were confirmed cases (blue bars), including 17 with equine squamous gastric disease (ESGD), 17 horses with equine glandular gastric disease (EGGD), and 21 with both ESGD and EGGD. The remaining 25 horses showed no evidence of EGUS and served as controls (yellow bars).

TABLE 4 | Prevalences and predictive values (and 95% confidence intervals) of owner-reported clinical signs ($n = 80$) and fecal occult blood test results ($n = 49$) at presentation in 80 competition dressage horses with suspected equine gastric ulcer syndrome (EGUS), including 55 cases with equine squamous gastric disease (ESGD) and/or equine glandular gastric disease (EGGD), and 25 controls without evidence of EGUS.

	Cases				
	ESGD	**EGGD**	**ESGD/EGGD**	**Total cases**	**Controls**
Clinical signs	($n = 17$)	($n = 17$)	($n = 21$)	($n = 55$)	($n = 25$)
Girth aversion (n)	12	14	16	42	20
Prevalence	70.6% (46.9%–86.7%)	82.4% (59.0%–93.8%)	76.2% (54.9%–89.4%)	76.4% (63.7%–85.6%)	80.0% (60.9%–91.1%)
Sensitivity	0.71 (0.47–0.87)	0.82 (0.59–0.94)	0.76 (0.55–0.89)	0.76 (0.64–0.86)	
Specificity	0.20 (0.09–0.39)	0.20 (0.09–0.39)	0.20 (0.09–0.39)	0.20 (0.09–0.39)	
Odds ratio (p)	0.88 (0.71)	1.03 (> 0.99)	0.95 (> 0.99)	0.95 (0.78)	
Poor performance (n)	5	5	6	16	10
Prevalence	29.4% (13.3%–53.1%)	29.4% (13.3%–53.1%)	28.6% (13.8%–50.0%)	29.1% (18.8%–42.1%)	40.0% (23.4%–59.3%)
Sensitivity	0.29 (0.13–0.53)	0.29 (0.13–0.53)	0.29 (0.14–0.50)	0.29 (0.19–0.42)	
Specificity	0.60 (0.41–0.77)	0.60 (0.41–0.77)	0.60 (0.41–0.77)	0.60 (0.41–0.77)	
Odds ratio (p)	0.74 (0.53)	0.74 (0.53)	0.71 (0.54)	0.73 (0.44)	
Weight loss (n)	8	6	4	18	6
Prevalence	47.1% (26.2%–69.0%)	35.3% (17.3%–58.7%)	19.0% (7.7%–40.0%)	32.7% (21.8%–45.9%)	24.0% (11.5%–43.4%)
Sensitivity	0.47 (0.26–0.69)	0.35 (0.17–0.59)	0.19 (0.08–0.40)	0.33 (0.22–0.46)	
Specificity	0.76 (0.57–0.89)	0.76 (0.57–0.89)	0.76 (0.57–0.89)	0.76 (0.57–0.89)	
Odds ratio (p)	1.96 (0.18)	1.47 (0.50)	0.79 (0.73)	1.36 (0.60)	
Behavioral changes (n)	5	4	4	13	9
Prevalence	29.4% (13.3%–53.1%)	23.5% (9.6%–47.3%)	19.0% (7.7%–40.0%)	23.6% (14.4%–36.3%)	36.0% (20.2%–55.5%)
Sensitivity	0.29 (0.13–0.53)	0.24 (0.10–0.47)	0.19 (0.08–0.40)	0.24 (0.14–0.36)	
Specificity	0.64 (0.45–0.80)	0.64 (0.45–0.80)	0.64 (0.45–0.80)	0.64 (0.45–0.80)	
Odds ratio (p)	0.82 (0.75)	0.65 (0.50)	0.53 (0.33)	0.66 (0.29)	
Poor coat condition (n)	4	4	5	13	5
Prevalence	23.5% (9.6%–47.3%)	23.5% (9.6%–47.3%)	23.8% (10.6%–45.1%)	23.6% (14.4%–36.3%)	20.0% (8.9%–39.1%)
Sensitivity	0.24 (0.10–0.47)	0.24 (0.10–0.47)	0.24 (0.11–0.45)	0.24 (0.14–0.36)	

(Continues)

TABLE 4 | (Continued)

	Cases				
	ESGD	**EGGD**	**ESGD/EGGD**	**Total cases**	**Controls**
Specificity	0.80 (0.61–0.91)	0.80 (0.61–0.91)	0.80 (0.61–0.91)	0.80 (0.61–0.91)	
Odds ratio (*p*)	1.18 (> 0.99)	1.18 (> 0.99)	0.96 (> 0.99)	1.18 (0.78)	
Chronic diarrhea (*n*)	3	3	4	10	7
Prevalence	17.6% (6.2%–41.0%)	17.6% (6.2%–41.0%)	19.0% (7.7%–40.0%)	18.2% (10.2%–30.3%)	28.0% (14.3%–47.6%)
Sensitivity	0.18 (0.06–0.41)	0.18 (0.06–0.41)	0.19 (0.08–0.40)	0.18 (0.10–0.30)	
Specificity	0.72 (0.52–0.86)	0.72 (0.52–0.86)	0.72 (0.52–0.86)	0.72 (0.52–0.86)	
Odds ratio (*p*)	0.63 (0.49)	0.63 (0.49)	0.68 (0.51)	0.65 (0.38)	
Bruxism (*n*)	1	3	4	8	7
Prevalence	5.9% (1.0%–27.0%)	17.6% (6.2%–41.0%)	19.0% (7.7%–40.0%)	14.5% (7.6%–26.2%)	28.0% (14.3%–47.6%)
Sensitivity	0.06 (0.00–0.27)	0.18 (0.06–0.41)	0.19 (0.08–0.40)	0.15 (0.08–0.26)	
Specificity	0.72 (0.52–0.86)	0.72 (0.52–0.86)	0.72 (0.52–0.86)	0.72 (0.52–0.86)	
Odds ratio (*p*)	0.21 (0.11)	0.63 (0.49)	0.68 (0.51)	0.52 (0.22)	
Recurrent colic (*n*)	2	2	1	5	5
Prevalence	11.8% (3.3%–34.3%)	11.8% (3.3%–34.3%)	4.8% (0.8%–22.7%)	9.1% (3.9%–19.6%)	20.0% (8.9%–39.1%)
Sensitivity	0.12 (0.02–0.34)	0.12 (0.02–0.34)	0.05 (0.0–0.23)	0.09 (0.04–0.20)	
Specificity	0.80 (0.61–0.91)	0.80 (0.61–0.91)	0.80 (0.61–0.91)	0.80 (0.61–0.91)	
Odds ratio (*p*)	0.58 (0.68)	0.58 (0.68)	0.24 (0.20)	0.45 (0.27)	
Fecal occult blood test ($n = 49$)	($n = 9$)	($n = 10$)	($n = 14$)	($n = 33$)	($n = 16$)
Albumin positive (*n*)	7	6	8	21	7
Prevalence	77.8% (45.3%–93.7%)	60.0% (31.3%–83.2%)	57.1% (32.6%–78.6%)	63.6% (46.6%–77.8%)	43.8% (23.1%–66.8%)
Sensitivity	0.78 (0.45–0.96)	0.60 (0.31–0.83)	0.57 (0.33–0.79)	0.64 (0.47–0.78)	
Specificity	0.56 (0.33–0.77)	0.56 (0.33–0.77)	0.56 (0.33–0.77)	0.56 (0.33–0.77)	
Odds ratio (*p*)	1.78 (0.21)	1.37 (0.69)	1.31 (0.72)	1.46 (0.23)	

(Continues)

TABLE 4 | (Continued)

	Cases				
	ESGD	**EGGD**	**ESGD/EGGD**	**Total cases**	**Controls**
Hemoglobin positive (*n*)	3	3	4	10	3
Prevalence	33.3% (12.1%–64.6%)	30.0% (10.8%–60.3%)	28.6% (11.7%–54.6%)	30.3% (17.4%–47.3%)	18.8% (6.6%–43.0%)
Sensitivity	0.33 (0.12–0.65)	0.30 (0.11–0.60)	0.29 (0.12–0.55)	0.30 (0.17–0.47)	
Specificity	0.81 (0.57–0.93)	0.81 (0.57–0.93)	0.81 (0.57–0.93)	0.81 (0.57–0.93)	
Odds ratio (*p*)	1.78 (0.63)	1.60 (0.64)	1.52 (0.67)	1.62 (0.50)	

Note: Predictive values are presented according to pathological classification in comparison with controls.

changes had both poor sensitivity and poor specificity for all disease classifications.

None of the ORs were significant ($p = 0.11–0.99$).

3.3.2 | After Treatment of EGUS

Of the seven cases that fully recovered from EGUS, five (71%; 95% CI: 36%–92%) showed no clinical signs at follow-up, as reported by the owners. Two horses (29%; 95% CI: 8%–64%) still exhibited clinical signs: one exhibited girth aversion, and the other had poor performance. According to the owners, of the 12 cases that did not fully recover from EGUS, seven (58%; 95% CI: 32%–81%) were free of clinical signs. The remaining five cases (42%; 95% CI: 19%–68%) exhibited girth aversion ($n = 4$), poor performance ($n = 2$), behavioral changes ($n = 2$), chronic diarrhea ($n = 1$), and recurrent colic ($n = 1$).

The frequency of clinical signs reported by owners after treatment is presented in Figure 3. Sixteen of the 19 horses (84%; 95% CI: 62%–95%) showed girth aversion before treatment. After treatment, five of these horses (31%; 95% CI: 12%–49%) still showed girth aversion (two with persistent EGGD, two with persistent ESGD/EGGD, and one that had recovered). Of the 11 horses in which girth aversion resolved after treatment, seven (64%; 95% CI: 35%–85%) still had EGUS (six with EGGD and one with ESGD/EGGD), while four (36%; 95% CI: 15%–65%) had fully recovered. Nine of the 19 horses (47%; 95% CI: 27%–68%) performed poorly before treatment. After treatment, three of these horses (33%; 95% CI: 12%–65%) still performed poorly (two with persistent ESGD/EGGD, and one that had recovered). Of the six horses in which poor performance resolved after treatment, four (67%; 95% CI: 30%–90%) still had EGGD, while two (33%; 95% CI: 10%–70%) had fully recovered. The frequency of the owner-reported clinical signs did not differ at follow-up examination compared to the initial presentation ($p = 0.08–0.99$).

3.4 | Predictive Value of FOBTs

Of the 49 horses tested with the FOBT, 28 (57%; 95% CI: 43%–70%) were positive for albumin, with no difference between cases and controls ($p = 0.23$; Table 4). Thirteen horses (27%; 95% CI: 16%–40%) tested positive for hemoglobin, with no difference between cases and controls ($p = 0.50$). Only one case (3%; 95% CI: 1%–15%) tested positive for hemoglobin while testing negative for albumin.

Detection of fecal albumin had moderate sensitivity for predicting ESGD but poor sensitivity for EGGD, ESGD/EGGD, and overall (all cases), as well as poor specificity. Fecal hemoglobin had poor sensitivity across all disease classifications but had acceptable specificity for EGUS. None of the ORs were significant ($p = 0.21–0.72$).

4 | Discussion

It is widely recognized that EGUS is prevalent in sports horses, presenting variable clinical manifestations and a considerable

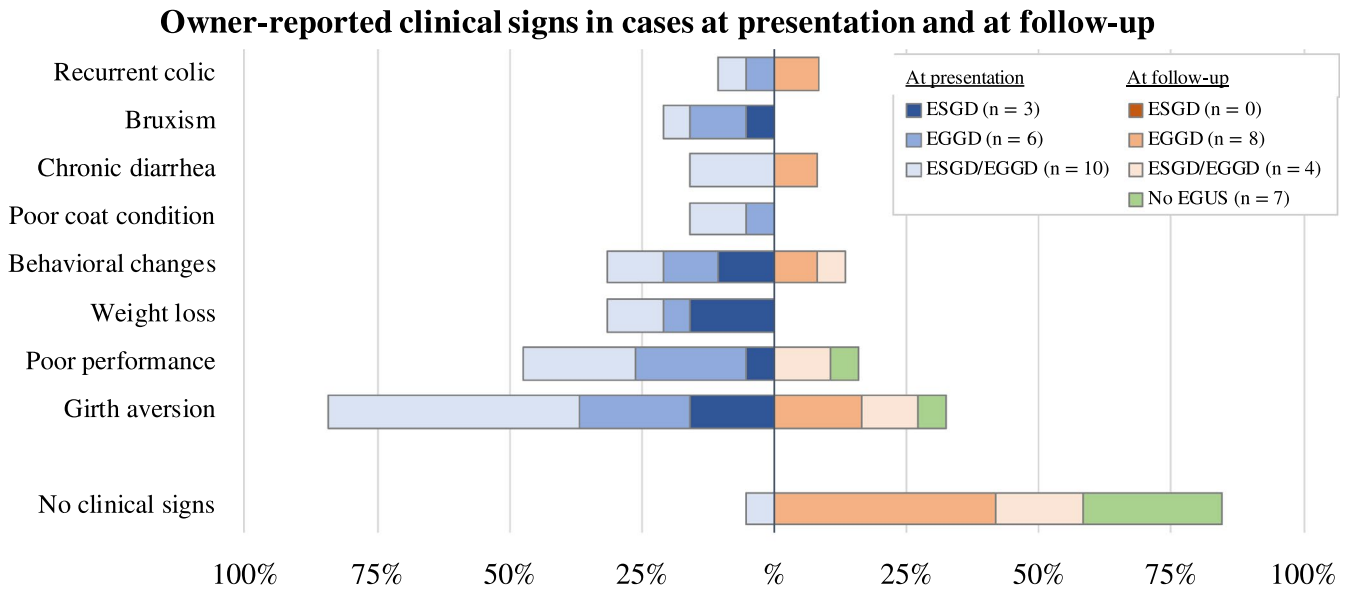

FIGURE 3 | Frequency of owner-reported clinical signs in 19 competition dressage horses with confirmed equine gastric ulcer syndrome (EGUS) at presentation (blue bars) and at follow-up (i.e., immediately after a course of treatment with oral omeprazole with or without sucralfate). At follow-up, 12 horses still had EGUS (orange bars), and 7 horses had recovered from EGUS (green bars). EGGD: equine glandular gastric disease; ESGD: equine squamous gastric disease.

effect on performance [2]. Accurate diagnosis is crucial to optimize welfare and athletic outcomes. In this referral cohort of actively competing Dutch dressage horses, EGUS was common (69%), yet neither owner-reported clinical signs nor FOBT predicted the syndrome. Girth aversion, frequently reported, was equally present in affected and unaffected horses. Thus, the limitations of subjective clinical signs and stall-side tests are highlighted.

Competition and training are risk factors for EGUS in sports horses. Ulcers often develop during or after competition. In one investigation, 47% of 23 previously unaffected sports horses developed EGUS after three consecutive competition days, although there was no differentiation between ESGD and EGGD [45]. Another study in 30 endurance horses reported ESGD prevalence rising from 48% off-season to 93% during competition, with an association with training and performance level [9]. In the same population, EGGD prevalence varied from 17% to 33%. In 101 breeding stallions, exercise was associated with EGGD (ORs 1.6–2.4), but not ESGD [53]. The high prevalence observed in our study likely reflects both actual disease burden and referral bias.

This study investigates EGUS prevalence in a referral cohort of active national or international competition dressage horses. Notably, 31% of horses referred for suspected EGUS had no confirmed ulcers, underscoring the essential role of gastroscopy in diagnosis and targeting appropriate treatment. Expert consensus discourages initiating treatment without gastroscopy to differentiate ESGD from EGGD and to avoid unnecessary medication [2].

Although no clinical sign is pathognomonic for EGUS [25], girth aversion and poor performance are commonly regarded as indicative within the equestrian sports community [20]. Girth aversion is linked to 25% of ESGD and 31% of EGGD cases, and poor performance to 44% and 49%, respectively [22, 23]. In the present study, girth aversion was reported in 76% of cases (74% of ESGD and 79% of EGGD) and 80% of controls. Poor performance was noted in 29% of cases (29% of ESGD and 29% of EGGD) and 40% of controls. This poor discriminative value highlights the limitations of relying on clinical signs alone for EGUS screening. Alternative causes of girth aversion and poor performance (e.g., orthopedic pain or ill-fitting saddles) were not investigated in this study because horses were explicitly referred for EGUS investigation. Only 32% of 37 sports horses with girth aversion had EGUS, while others had orthopedic problems (27%), ill-fitting saddles (8%), and abdominal conditions (30%) [24]. Abnormal behavior during tacking-up and mounting is common [54], and can be associated with an ill-fitting saddle, lameness, or epaxial muscle pain [55]. Poor performance is a nonspecific sign commonly associated with musculoskeletal, cardiac, or respiratory disorders, such as asthma or respiratory viruses [56–59]. In the present study, follow-up gastroscopy revealed no relationship between owner-reported clinical outcomes and mucosal healing: 71% of horses with healed ulcers were reported as improved, yet 58% with persistent ulcers had no clinical signs. This raises the possibility that acid suppression might temporarily alleviate clinical signs without achieving mucosal healing. However, follow-up data are subject to selection bias (particularly the likelihood that horses perceived as clinically recovered were less likely to return), limiting the generalizability of the results. Another study found behavioral or performance changes in only 24% of horses treated for EGUS with abnormal tacking-up behavior [54]. These findings highlight that clinical response alone is an unreliable marker for healing, underscoring the importance of follow-up gastroscopy.

The relationship between EGUS and clinical signs warrants closer scrutiny because signs commonly attributed to EGUS were also frequent in horses without endoscopic evidence of gastric lesions. This overlap suggests that EGUS might not be

the primary cause of these signs in all cases, risking misdiagnosis [60]. Some behavioral markers (e.g., frequent head turning toward the abdomen and deep abdominal stretches) might contribute to diagnostic assessment, although current evidence only documents their presence in horses with EGUS and does not confirm their specificity for the condition [61]. The inconsistent correlation between lesion severity and clinical signs further complicates establishing causality [3, 53, 62]. These findings highlight the need for more research to determine whether EGUS is truly responsible for signs like poor performance or if its role has been overstated. Additionally, clinical signs, especially pain-related ones, can be multifactorial, and conditions such as EGUS and musculoskeletal disorders can coexist, contributing additively to a horse's presentation. Therefore, a comprehensive clinical assessment beyond gastroscopy is essential for horses presenting with poor performance or nonspecific clinical signs.

The FOBT evaluated in this study is a lateral flow immunoassay detecting albumin and hemoglobin in feces. It demonstrated high in vitro analytical accuracy, with 100% sensitivity and specificity for albumin detection, and 100% sensitivity and 97% specificity for hemoglobin [36]. Albumin, released during gastrointestinal tissue injury, is enzymatically degraded in the small intestine, so fecal albumin likely originates from lesions in the hindgut. Hemoglobin, released during active bleeding, is only partially degraded, and fecal hemoglobin might indicate bleeding anywhere in the gastrointestinal tract. Manufacturer guidelines suggest that hemoglobin alone indicates foregut bleeding, while the combined presence of albumin and hemoglobin might reflect hindgut or concurrent lesions. In the present study, FOBT was performed in 49 horses due to limited kit availability. Although a manufacturer-reported *postmortem* study showed high analytical sensitivity and specificity for equine gastrointestinal lesions [41], the present study found the FOBT to have poor sensitivity for detecting EGUS, with inadequate performance across all disease classifications. This aligns with research reporting limited ability to accurately identify gastric lesions, regardless of location or severity [42, 43, 63]. Poor sensitivity implies a substantial risk of false negative results, meaning many affected horses would remain undiagnosed if relying on FOBT alone. Consequently, FOBT lacks clinical utility as a screening tool for EGUS in sport horses [64].

Treatment followed European College of Equine Internal Medicine consensus recommendations, using omeprazole alone for ESGD and omeprazole with sucralfate for EGGD or combined disease [2]. No horses received antimicrobials, aligning with antimicrobial stewardship [2, 65]. Treatment outcomes mirrored prior findings, with higher healing rates for ESGD than EGGD [66–68], likely reflecting differences in pathogenesis and acid suppression response [60]. Environmental and management improvements (e.g., increasing turnout, social interaction, and varied exercise) enhance healing [49, 69]. However, sports horses are typically managed within highly structured professional programs, and practical constraints in these competitive environments (e.g., logistical, economic, and competitive demands) often limit their implementation [1]. Although based on a subset of horses returning for follow-up and potentially subject to selection bias, these findings offer clinically relevant insights into challenges encountered in EGUS therapy in sports horses.

Several limitations should be noted. Clinical signs were owner-reported, consistent with the aim to evaluate their predictive value, which may introduce potential bias in reporting. Horses were explicitly referred for EGUS evaluation, and investigation was limited to clinical examination, gastroscopy, and FOBT. Investigating poor performance is particularly complex because there are many potential causes [70]. No lameness evaluation, dental assessments, or standardized exercise tests were conducted [71], limiting the detection of comorbidities. Exercise-related discomfort was not assessed using the Ridden Horse Pain Ethogram [35, 72], because of the need to withhold food and water before gastroscopy. Only 35% of cases underwent follow-up gastroscopy, introducing potential selection bias, because owners of clinically improved horses might have declined reexamination. Long-term effects of treatment on EGUS and clinical signs remain unknown. The effect of withholding food for up to 16 h on FOBT remains uncertain. While temporary starvation affects albumin synthesis in rabbits [73], increases serum hemoglobin in monkeys [74], and alters intestinal mucosa in humans [75], its influence on FOBT in horses has not been studied.

In conclusion, while EGUS is highly prevalent in sports horses, the findings of the present study indicate that owner-reported clinical signs and FOBT were unreliable for diagnosis, emphasizing the importance of gastroscopy in suspected cases. Empirical treatment would have been unnecessary in 31% of horses exhibiting clinical signs, suggesting alternate diagnoses. Follow-up gastroscopy is recommended, as many clinically improved horses still had gastric lesions, and over a quarter with ongoing clinical problems had healed ulcers. Gastroscopy remains the reference standard for the diagnosis and monitoring of EGUS. These findings reinforce the need for a deeper understanding of EGUS pathogenesis and its role in equine performance.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

All procedures in this study complied with both national and international guidelines for animal welfare. The Animal Ethics Committee of the University of Liège was consulted (Approval No. 24–2765) and confirmed that the methods used in this study are part of routine veterinary practice for diagnosing EGUS. Consequently, formal ethical approval was not required. Informed consent was obtained from all horse owners before their inclusion in the study. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **File S1:** Standardized form for recording owner-reported history, body condition score (9-point scale), training/competition status, medical history, diet, management, clinical examination findings, gastroscopy results, and fecal occult blood test outcomes in studied horses. **File S2:** Manufacturer's instructions for the SUCCEED Equine Fecal Blood Test, a qualitative lateral flow immunoassay detecting albumin and hemoglobin in equine feces. The document details patient considerations, sample collection, test preparation, application of diluted fecal material to the cassette wells, and interpretation of results.